INCH-POUND

A-A-59340B

13 December 2006

SUPERSEDING

A-A-59340A

25 Sep 2000

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 10,000 POUND CAPACITY

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This commercial item description covers the general requirements for a diesel engine driven, front wheel drive, rear wheel steer, pneumatic tired, 10,000 pound capacity, and forklift truck. The forklift is intended for handling cargo on the 463L pallet and miscellaneous cargo in and around aircraft and warehouses, loading platforms, docks, and on paved or other hard surfaces.

2. CLASSIFICATION. The forklift shall be one of the following types and classes. The type and class to be furnished shall be as specified (see 7.2).

Type I:	10,000 pound capacity at 48 inch load center
Type II:	10,000 pound capacity at 24 inch load center
Class A:	Nuclear Certified
Class B:	Non-Nuclear Certified

3. SALIENT CHARACTERISTICS:

3.1 Safety. The forklift shall comply with ASME/ANSI B56.1, UL 558, and OSHA standards in effect at the time of manufacture. The vehicle shall be DS rated. An operator's seatbelt conforming to SAE J 386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tip over shall be provided.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: WR-ALC/GBZO-ENV, 380 Richard Ray Blvd Ste 104, Robins AFB GA 31098-1640

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3.1.1 Design and Safety (Class A). In addition to the requirements of 3.1, the forklift shall also meet the requirements of Air Force Manual 91.1 18. The forklift shall be designed for a structural safety factor (based on yield strength) of at least 3 times the rated load, or maximum dynamic load multiplied by a factor of 2, whichever is greater. The forklift shall be designed to minimize the transmission of mechanical shock to loads in all modes of operation.

3.2 Environmental requirements. The truck shall be capable of operating in temperatures ranging from 0 degrees F to +125 degrees F. When specified (see 7.2) the truck shall be equipped with winterization A or C.

3.2.1 Type A winterization. When specified (see 7.2), the truck shall be furnished with Type A winterization that shall protect to -65 degrees F and shall consist of the following:

a. Power plant heaters. Engine coolant, engine oil and battery heaters shall be provided. All heaters shall operate on either 110 volts at 60 Hz or 220 volts at 50 Hz alternating current. A three wire, 25 feet long weatherproof cable of adequate capacity for all heaters being used simultaneously shall be provided. The cable shall be able to connect to NATO and American connectors. Stowage shall be provided on the truck to keep the cable when not in use. The heaters shall be as follows:

    1. The coolant heater shall be installed in the engine block or lower coolant inlet hose. A coolant circulating pump, driven by a 110 volt or 220 volt alternating current motor shall be provided when a coolant inlet hose heater is furnished. The heater shall have adequate capacity to maintain engine coolant at a temperature of 10 degrees F in an ambient temperature of -40 degrees F. It shall be controlled to limit engine coolant to not more than 150 degrees F.

    2. An engine oil heater with adequate capacity to maintain engine oil at a temperature of at least +10 degrees F in an ambient temperature of -40 degrees F shall be furnished. It shall be controlled to limit engine oil temperature to not more than 150 degrees F.

    3. A battery heater shall be provided. It shall have adequate capacity to maintain battery electrolyte at a temperature of at least +10 degrees F in an ambient temperature of -40 degrees F, and shall have a thermostat to limit temperature of the electrolyte to not more than +80 degrees F.

b. Cab. The cab shall meet the requirements of 3.21, and shall also contain thermal insulation, if necessary, to meet heating requirements mentioned below.

c. Heater and defroster. The heater shall be of sufficient capacity to maintain a temperature of +40 degrees F at cab floor level in an ambient temperature of -40 degrees F

d. Cold starting aid. A measured shot ether system or glow plug(s) shall be provided to assist in engine cold temperature starting.

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e. Anti-freeze engine. To protect the engine from freezing, the coolant system shall be protected to -65 degrees F with antifreeze.

3.2.2 Type C winterization. When specified (see 7.2), the truck shall be furnished with Type C winterization that shall protect to -25 degrees F and shall consist of the following:

a. Starting aid. Glow plug(s) or a measure shot ether injection system shall be furnished.

b. Cab. The cab shall meet the requirements 3.21, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 3.22.

c. Heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least +40 degrees F at cab floor level in an ambient temperature of -20 degrees F.

d. Antifreeze. Engine coolant shall be protected to -25 degrees F with antifreeze.

3.3 Engine. The forklift shall be powered by a standard, commercial, diesel engine. The engine shall operate on diesel fuel in accordance with A-A-52557 without detrimental effects on the engine, or its performance.

3.4 Fuel system. The forklift shall have all necessary filters, water separators, and components required for diesel fuel operation. The forklift shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

3.5 Starting system. The starter switch shall not activate the engine starter while the engine is running, nor when the engine is not running and the transmission is in any forward or reverse gear. The starter switch shall only operate when the transmission is in the “neutral” or “park” positions.

3.6 Transmission. A torque converter and a power shift type of transmission shall be furnished, with at least two speeds forward and two speeds reverse. The transmission shall provide for positive inching control or de-clutch control that permits lifting of rated load at maximum lifting speed while the transmission control is in either forward or reverse direction, without vehicle movement. This control shall also apply the service brakes. When specified (see 7.2), a transmission oil cooler shall be furnished.

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3.7 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system. The hydraulic system shall also be designed to prevent forward tilt in the event of system failure; tilt angle shall not deviate more than two degrees if failure occurs at any position during load handling operations.

a. Class A. The hydraulic lift system shall be designed so that in the event of hydraulic hose or pump failure, the lift system will automatically prevent load lowering. A manual override for load lowering and tilt functions shall also be provided to safely lower the load and provide for load removal; lowering speed of the forks shall not exceed eight (8) inches per second under any load conditions.

b. Class B. The hydraulic lift system shall be designed so that in the event of hydraulic hose or pump failure, the load shall not descend (freefall) at an uncontrolled rate; lowering speed shall not exceed eight (8) inches per second under any load condition. A manual override for the tilt function shall also be provided to allow for load removal.

3.8 Uprights and carriage. With no load on the forks, the seated operator with seatbelt engaged shall be able to see at least one fork tip at all lift heights and fork positions.

3.9 Forks and carriage. Fork tines shall have the following dimensions. The forks shall be mounted so that when fully lowered and the forklift is operating in reverse, there is no damage or degradation to the fork tines, their attachment, or any other component of the carriage, including fork positioner and side shift.

Type I:

a. Length: 72 ± 1 inches

b. Width: 8 inches (maximum)

c. Thickness: 2.5 inches maximum, except bend in fork heel may exceed 2.5 inches.

d. Fork tip thickness: ½ inch (maximum)

e. Fork taper (underside of forks): Taper from fork tip rearward shall be 30 inches (minimum).

f. Fork carrier assembly width shall not exceed the overall width of the forklift truck.

Type II: Fork tines shall be 48 inches in length with an underside taper. When Specified (see 7.2), the fork tines shall be 72 inches in length with an underside taper. All fork tip thickness shall not exceed ½ inches.

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3.10 Fork positioner. A hydraulically powered fork positioner shall be furnished and controlled by the operator while seated. The fork positioner shall allow tine spacing of 18 inches (or less) to 60 inches (or more), measured between centerlines of the forks.

3.11 Side shift. Hydraulically powered side shift capability shall be furnished. It shall be controlled by the seated operator, and shall provide at least six inch side shift capability (each side of center or 12.0 inches minimum total travel).

3.12 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 48 inches high, measured from the load carrying surfaces of the forks.

3.12.1 Load backrest restraints (Class A). Two ring type devices to restrain the load shall be located on the load backrest approximately 30 inches above the load carrying surfaces of the forks. The devices shall have a minimum inside diameter of two inches. The devices shall have a structural safety factor of at least 3:1 when a forward horizontal load, equal to the rated load, is equally applied to both devices. Location of the devices shall not interfere with positioning the load against the backrest.

3.13 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

3.14 Service brakes. Power brakes conforming to ASME/ANSI B56.1 shall be furnished, except maximum brake pedal force during stopping shall not exceed 80 pounds.

3.14.1 Service brakes (Class A). The service brakes shall also be capable of holding the forklift, with rated load, on a minimum 11.5 degree grade without assistance of the parking brake.

3.15 Parking brake. Parking brake shall conform to ASME/ANSI B56.1.

3.15.1 Parking brake (Class A). In addition to the requirements of 2.15, the parking brake shall be independent of the service brake system and shall be capable of holding the forklift, with rated load, on a minimum 11.5 degree grade.

3.16 Electrical system. A 12 volt direct current (VDC) electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

3.16.1 Backup alarm. An audible backup alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse.

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3.17 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter and tachometer shall be installed in the instrument panel. All instruments shall illuminate, except for the hour meter. All load motion controls shall be right hand operation controls and of the self-centering type, i.e. controls shall return to the neutral position when released.

3.18 Lighting. Minimum four sealed beam floodlights, two in front facing forward and two mounted in the rear facing rearward, shall be furnished. An additional floodlight shall be furnished that is adjustable from inside the cab by the operator while seated; the floodlight shall provide at least 270" rotation capability. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location or guards. Individual operator controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

3.19 Horn. Horn button shall be mounted in the center of the steering wheel.

3.20 Operator's overhead guard. The guard shall be in accordance with ASME/ANSI B56.1. It shall also be a falling-object-protective-structure (FOPS) conforming to ASME/ANSI B56.1. The overhead guard height from ground to top of guard shall be 102 inches maximum.

3.21 Cab. The forklift shall be equipped with an enclosed cab that shall incorporate overhead protection requirements of ASME/ANSI B56.1. The cab shall have a door on each side of the vehicle with at least one window per door which can be opened for ventilation; the windows shall be capable of being locked in either the open or closed positions. The doors shall be provided with hinges that can be serviced without door removal and shall allow adjustments for door fit. The doors shall be capable of being locked in either the opened or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and roof window. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant non-absorbent thermal insulation. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2 - ½ pound capacity ABC type, or equivalent. The fire extinguisher shall be UL listed.

3.22 Heater and defroster. The manufacturer's heaviest duty heater and defroster shall be furnished.

3.23 Windshield wipers. There shall be at least three wipers furnished, one on the front windshield, one on the rear window, and one on the roof window.

3.24 Performance.

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3.24.1 Type I:

a. Capability to safely handle a 10,000 pound load, with a 48 inch load center, at all lift heights

b. Lifting speed. At least 14 inches per second, but not to exceed 17 inches per second.

c. Lowering speed. Not more than 14 inches per second with rated load on forks. Not less than eight (8) inches per second with rated load and with engine at idle speed.

d. Upright height with forks on ground (collapsed mast height): 97 inches maximum, with no load on the forks.

e. Maximum fork lift height. At least 150 inches with rated load on forks and forks horizontal.

3.24.2 Type II:

a. Capability to safely handle a 10,000 pound load, with a 24 inch load center, at all lifts heights.

b. Lifting speed. At least 11 inches per second.

c. Lowering speed. Not more than 16 inches per second with rated load on the forks. Not less than 8 inches per second with rated load and with engine at idle speed.

d. Upright height with forks on ground (collapsed mast height): 130 inches maximum, with no load on the forks.

e. Maximum fork lift height. At least 168 inches with rated load on forks and forks horizontal.

3.24.3 Type I and II.

a. Right angle turn. With rated load on the forks, forklift positioned perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 210 inches.

b. Travel speed. 12 miles per hour (mph) minimum (not to exceed 18 mph), in both forward and reverse, while carrying rated load on the forks. The vehicle speed shall be governed so that maximum speed for safe operations is not exceeded.

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c. Slope ascension. With the truck facing up the slope and with rated load on the forks, the truck shall be able to accelerate up a 20 percent slope from a dead stop.

d. Upright tilt. With no load on the truck, the forks shall have at least 3.0 degrees of forward tilt, and at least 6.0 degrees of rear tilt.

e. Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 5.0 inches. With no load on the forks, the ground clearance of the truck other than the mast and axles, shall be at least 7.0 inches.

f. Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height for at least 2.0 minutes with not more than 1- ¾ inch of vertical drift, and not more than 1.0 degree of rotational drift.

g. Stability. The forklift shall meet the ASME/ANSI B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.

h. Noise limits. Maximum allowable noise level shall not exceed 84 dB(A) at the operator's ear. If the noise level at the operator's ear exceeds 84 dB(A) (85 to 92 dB(A)), a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level limits that exceed 92 dB(A) are not acceptable.

3.25 Painting. The forklift shall be primed and painted using standard commercial practices. Unless otherwise specified (see 7.2):

a. Type I: Color shall be 24052 of FED-STD-595.

b. Type II: Manufacturer's standard color shall be furnished.

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3.26 Markings, data plates, and instruction plates. Corrosion resistant identification plate, hydraulic schematic, wiring schematic, and all warning and instruction plates shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

Class A:	Class B:
Truck, Lift, Fork	Truck, Lift, Fork
Capacity: 10,000 pounds at (_) inch load center	Capacity: 10,000 pounds at (_) inch load center
Gross Vehicle Weight (GVW):	Gross Vehicle Weight (GVW):
National Stock Number (NSN):	National Stock Number (NSN):
Contract Number:	Contract Number:
Serial Number:	Serial Number:
Model Number:	Model Number:
Registration Number:	Registration Number:
Manufactured by:	Manufactured by:
Name:	Name:
Address:	Address:
Service Phone Number:	Service Phone Number:
US Property:	US Property:
ANSI B56.1 conformance:	
UL Type designation and conformance:	

3.27 Walkway coating. Floor plates and step surfaces shall be coated with a non-slip coating compound, or be furnished with a non-slip metal or tread plate surface.

3.28 Lifting and tie-down provisions. There shall be permanently installed lifting devices that enable the forklift to be lifted in its normal travel configuration, and tie-down devices that enable the forklift to be tied down to the floor of the transportation media. All devices shall have an inside diameter of 3.0 inches and minimum structural safety factor of 2:1, based on static load. The notation "LIFT HERE" and "TIE DOWN" shall be stenciled, in black, near each lifting and tie-down device.

3.29 Electromagnetic interference (EMI). The forklift shall comply with EM1 requirements of MIL-STD-461 RE 102 and RS 103.

3.30 Highway transportability. The truck, when loaded on a semi trailer, shall be within the highway permit limits for all states.

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3.31 Air transportability (Type I). The forklift shall be designed for air transport in C-130, C-141, C-5 and C-17 aircraft, refer to MIL-HDBK-1791 as a guide. The forklift shall be capable of being driven on and off the aircraft and shall not exceed maximum axle weight requirements without removal of any component of the vehicle. The forklift shall be furnished with sufficient tie-down provisions for securing the vehicle for air transport. All tie-down locations shall be properly marked and maximum capacity indicated (see 3.28). The center of gravity (CG) of the forklift shall be marked on both sides of the vehicle. The manuals shall list the dimension from the front axle rearward to the CG.

3.32 Truck operations. The truck shall complete not less than 3 circuits per hour, for a cumulative total of 135 hours, when tested as specified in (see 5.3.22). Only routine maintenance as prescribed by the manufacturer is allowed and only at the end of each operational period. The truck shall show no signs of unusual or detrimental wear. No failure shall occur during the 135 hour operational test. A failure is defined as any isolated malfunction that cannot be corrected within 30 minutes by adjustment, repair or replacement, which may cause cessation of operation, degradation of performance capabilities below specified levels, damage to the truck by continued operation, or personnel safety hazard. Note: More than one malfunction of an item shall be considered a failure.

4. REGULATORY REQUIREMENTS

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift. The forklift shall comply with applicable Environmental Protection Agency (EPA) regulations.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.1.1 Class A. Unless otherwise specified (see 7.2) the contractor is responsible for the performance of all inspections and testing specified herein, at his own or other facilities of his choosing. Government representatives shall witness all testing specified in Section 5 of this document.

5.2 Responsibility for compliance (Class A). All trucks on the contract shall meet all requirements of this document. The contractor shall comply with all requirements of this document, whether or not there is a corresponding inspection requirement in Section 5 of this document.

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5.3 First article inspection (Class A). The first article truck and its components, contractor drawings and literature, and vendor literature shall be compared to Section 5 of this document to verify compliance. The truck shall then be subjected to the demonstrations and test specified in 5.3.1 through 5.3.22.

5.3.1 Demonstrations and test (Class A). The following demonstrations and tests may be performed in any order, as determined by the contractor. The forklift shall be examined to determine compliance with this document. Occurrence of any of the following conditions during the testing specified herein shall be cause for rejection of the truck:

a. Inability to meet specified performance requirements or inability to complete any test.

b. Permanent deformation, over-heating, malfunctions, leakage of water, transmission fluid, engine oil, fuel, lubrications, hydraulic fluid, etc, in excess of Class 3, for dust free conditions and Class 3D for dusty conditions in accordance with SAE J1176.

c. Any binding, jerking, or uneven operation of any component or system.

d. Any wearing, galling, gouging or other material removal.

e. Any unsafe operation that jeopardizes the operator or forklift.

5.3.2 Demonstration conditions (Class A). Unless otherwise specified (see 7.2), demonstrations shall be conducted at the ambient temperature and climatic conditions existing at the place of demonstration. Only that maintenance established by the contractor shall be performed during the demonstration. All distance and time measurements, and other required data obtained during the testing shall be recorded and included in the first article test report.

5.3.3 Rated load. The rated load shall be a 10,000 pound (+200, -0) pound cube with a center of gravity located at the geometrical center of the cube. The cube shall be sized to provide a 48 inch load center on the forks of type I vehicles. The cube shall be sized to provide a 24 inch load center on the forks of type II vehicles. For travel and service brake testing, the 10,000 pound load can be dimensionally altered to provide visibility for the operator.

5.3.4 Maximum fork lift height. With rated load and forks level, raise forks to their maximum lift height. Measure from ground to top surface of one fork and determine maximum lift height. Failure to obtain a minimum 150 inch lift height for type I vehicles or a minimum 168 inch lift height for type II vehicles shall be cause for rejection.

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5.3.5 Lifting and lowering speed (Class A). Measure distance from ground to top surface of forks in fully lowered position. Measure time required to raise rated load from ground to maximum lift height. Subtract distance from ground to top surface of forks from maximum lift height, and calculate lifting speed. Lower rated load, at maximum lowering speed (fully open lowering control) and abruptly stop the load at 1 to 2 foot height. Record the time. Measure distance from ground to top surface of one fork. Subtract this figure from the maximum lift height and calculate loaded lowering speed. Remove load and raise forks to maximum lift height. Measure time required to lower forks to ground level. Calculate the unloaded fork lowering speed. Nonconformance with Section 3.24 shall be cause for rejection.

5.3.6 Travel speed (Class A). Operate the truck at maximum speed in both forward and reverse directions with rated load in load carry position. Inability to safely obtain 12 mph or exceeding 18 mph shall be cause for rejection.

5.3.7 Longitudinal slope operation (Class A). Drive the truck forward with rated load in uphill position onto a ramp, hill, or other suitable surface with a minimum 20 percent slope. Stop the truck on the slope, and then accelerate up the slope reaching a steady measured speed. The truck shall descend the slope, with the load in the uphill position, at a safe, measured speed.

5.3.8 Drift (Class A). Raise rated load to maximum lift height, with hydraulic fluid at ambient temperature. Place forks in horizontal position and shut off engine. Measure from ground to top surface of one fork. Let truck sit for a minimum 2.0 minutes. Using the same fork and reference points, measure from ground to top surface of fork. Vertical drift exceeding 1-3/4 inches, or rotational drift exceeding 1.0 degrees, shall be cause for rejection.

5.3.9 Right angle turn (Class A). With rated load on forks, position the truck so that it is perpendicular to a wall with the load facing the wall. Using the wall as a reference, measure a point 210 inches perpendicular from the wall. Mark location and other points 210 inches away from the wall so that a line parallel to the wall can be formed. Back vehicle so that a full right turn can be accomplished to the point the forklift is positioned parallel to the wall. Any portion of the forklift or load that extends beyond the 210 inch line shall be cause for rejection. Repeat test for left turn.

5.3.10 Fork tine operation (Class A). Measure both forks for length, width, thickness, thickness at tip, and bottom taper. With no load, move forks from minimum to maximum fork spread, and back to minimum position. Repeat at least five times. Demonstrate side-shift capability through its full range. Repeat at least five times each for unloaded and at rated capacity. With rated load on forks, raise forks to maximum lift height, then lower at maximum lowering speed and quickly stop at one to two feet above ground level. Repeat at least five times. Occurrence of any condition listed in 5.3 shall be cause for rejection. Position unloaded forks approximately horizontal and 3 to 5 inches above ground level. Measure the fork tilt. Without touching the tilt control lever, raise forks to maximum fork height, then lower to original position. Verify that the forks maintain the same tilt while being raised and lowered. Repeat procedure with rated load on forks and forks at approximately five degrees rear tilt. Inability to maintain the fork tilt shall be cause for rejection.

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5.3.11 Stability (Class A). Conduct the “longitudinal stability stacking”, “longitudinal stability travel”, “lateral stability stacking”, and “lateral stability travel” tilting platform tests of ASME/ANSI B56.1. Failure to meet the minimum tilting requirements specified in ASME/ANSI B56.1 shall be cause for rejection.

5.3.12 Electromagnet interference (EMI) (Class A). Test the truck to determine compliance with MIL-STD-461 RE 102 and RS 103. Failure to meet requirements shall be cause for rejection.

5.3.13 Service brakes (Class A). Demonstrate that the service brakes can stop the truck, with rated load, in accordance with ASME/ANSI B56.1. Position forklift, with rated load, on a minimum 11.5 degree incline and demonstrate the service brake system can hold the truck. Maintain position for one minute without assistance from the parking brake system. Failure to meet ASME/ANSI B56.1 requirements, or inability to maintain the forklift position on the incline shall be cause for rejection.

5.3.14 Parking brake (Class A). Demonstrate that the parking brake can hold the truck, with rated load, on a minimum 15 degree slope without assistance from the service brake system. Maintain position for at least three (3) minutes. Inability to maintain the forklift position shall be cause for rejection.

5.3.15 Starter disconnect switch (Class A). Start the engine and let it run for at least 10 seconds. Energize the starter switch (move to “off” then “on” position if applicable) while the engine is running. Any evidence of starter engagement while the engine is running shall be cause for rejection. With the engine not running, place transmission selector in the lowest forward gear and energize the starter switch. Repeat for all forward and reverse gears. Any evidence of starter engagement while transmission is in any gear shall be cause for rejection.

5.3.16 Overload (Class A). Perform a stress analysis of the truck to determine all critical stress points. Apply strain gauges or other suitable instrumentation at all the critical stress points to measure stress during the test. Place truck on a level surface and position forks at least 24 inches above ground level with the forks centered on the carriage. Position supports under the frame or front axle to relieve load on front tires, and secure the rear of the truck to compensate for the additional test load. For the test load, use the greater of 3 times the rated load or 2 times the maximum dynamic load, whichever is greater. Apply this test load to the forks. Let overload remain on the forks for at least 10 minutes. Remove the load and inspect truck structure, frame, mast, and hydraulic system for deformation, cracks, broken welds, hydraulic fluid leaks, etc. Inspect forks a minimum of six (6) inches on either side of heel by magnetic particle or dye penetrant method. Record readings from strain gauges and convert to, and record, the corresponding stresses. Failure to meet these requirements shall be cause for rejection. Include results of the stress analysis that determined the stress caused by the static and dynamic loads in the first article test report. The entire stress analysis is not necessary, just the location of critical stress points.

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5.3.17 Fail safe hydraulic lift system (Class A). Raise the rated load to maximum lift height with forks level. Measure vertical distance from ground to top surface of one fork. No personnel shall be under or in front of the load during the next step. Disconnect hydraulic line(s) that supply pressure to the lift system. Re-measure vertical distance from ground to same point on fork surface. A drop of more than 1/2 inch shall be cause for rejection. Measure fork tilt. Disconnect hydraulic line(s) that supply pressure to prevent forward tilting. Remeasure fork tilt. A change of more than 1.0 degrees of tilt shall be cause for rejection. Use the manual override provided to lower rated load to ground.

5.3.18 Fork lifting mechanism (Class A). Raise the rated load from ground level to maximum lift height and lower back to ground. This shall be considered one cycle. Repeat until 100 cycles have been completed. This is a severe test of the hydraulic system so raising and lowering may be stopped after every 10 cycles to allow cooling of the hydraulic system. Inability to complete the test, binding or erratic movement, any failure of the lifting system, or overheating of the hydraulic fluid shall be cause for rejection.

5.3.19 Fork spacing and side shift mechanism (Class A). The forks on rigid frame trucks shall be side shifted with the rated load from extreme left to extreme right position, and back to extreme left. This shall be considered one cycle. Repeat until 100 cycles have been completed. This is an extreme test of the hydraulic system and side shifting may be stopped after 20 cycles to allow cooling of the hydraulic fluid. For both rigid or articulated frame trucks, with no load on the forks, the forks shall be adjusted from their most closed position to most open position and then returned to most closed. This shall be considered one cycle. Repeat until 100 cycles have been completed. For both side shifting and fork spacing, any binding, erratic movements, inability to complete either test, failure of either system, or overheating of the hydraulic fluid shall be cause for rejection.

5.3.20 Air transportability verification (Type I). Verify the forklift can meet all air transportability requirements, including tie-downs, negotiation of ramps, and dimensional and weight limitations. Inability to meet requirements shall be cause for rejection. Refer to MIL-HDBK-1791 as a Guide.

5.3.21 Load backrest restraints (Class A). Apply a forward, horizontal force of 3.0 times rated load to load restraint devices for 10 minutes. The test load may be applied to all devices at once, or 3.0 times the proportional share of the load may be applied to each device. Remove load and inspect restraint devices, welds and supporting structure for deformation, cracks, broken welds, etc, which shall be cause for rejection.

5.3.22 Truck operations (Class A). The truck shall be driven at least 1.0 miles at varying speeds, including maximum speed. At least five right and five left turns shall be made at maximum turn angle. During the test, the horn, windshield wipers, lights, heater, defroster, and other components shall be operated. Occurrence of any conditions listed in 5.3, or in-operability of any component listed, shall be cause for rejection. Perform Test Method No. 1 (see figure 1), Truck Operations. Nonconformance with (see 3.36.16 or 5.3.1) shall constitute failure of this test.

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6. PACKAGING

6.1. Preservation, Packaging, Labeling and Marking. Unless otherwise specified (see 7.2), the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc. This involves shipment from the supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling and marking shall be as specified in the contract and/or order.

7. NOTES

7.1 Sources of referenced documents.

Department of Defense and Federal documents referenced herein may be obtained from the Document Automation and Production Service, Bldg 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094, or online at http://assist.daps.dla.mil .

The Society of Automotive Engineers (SAE) documents may be obtained online at http://www.sae.org/ or from SAE, 400 Commonwealth Drive, Warrendale, PA 15096.

ANSI/ASME documents may be obtained online from http://www.asme.org/ or from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

OCCUPATIONAL SAFETY AND HEALTH ASSOCIATION (OSHA)
OSHA Standards Occupational Safety and Health Standards for Industry
(Applications for copies should be sent to the U. S. Department of Labor, 200 Avenue NW, Room 423, Washington DC 20210), or online at http://www.osha.gov

Federal Acquisition Regulation (FAR) documents may be obtained at www.acqnet.gov/far or at the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250.

Underwriters Laboratories (UL) Industrial Trucks, Internal-Combustion Engine Powered (Applications for copies should be sent to the Underwriters Laboratories INC., 333 Pfingsten Road Northbrook, IL 60062), or online at http://www.ul.com

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7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type of Fork Truck required (see 2).
c. Winterization (see 3.2).
d. Transmission oil cooler, if required (see 3.6).
e. Fork dimensions (see 3.9).
f. Fork Lift primed and painted, if different (see 3.25).
g. Responsibility for performance, if different (see 5.1.1).
h. Demonstrations, if different (see 5.3.2).
i. Packaging requirements, if different (see 6.1).

7.3 Subject term (key word) listing.

1. 100000 Pound
2. Diesel-Engine-Driven
3. Forklift
4. Pneumatic-tired

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TEST METHODS NO. 1

TRUCK OPERATIONS

1. Demonstration course:

(a) The test course shall be a level, improved surface laid out in accordance with Figure A-1. Station 1 and Station 3 shall be platforms that are not less than 8 feet in height and capable of supporting rated load. Station 2 shall be at ground level.

(b) Pick up load from Station 1; back up truck being sure all wheels travel over obstacles; travel forward to Station 2. Before reaching Station 2, stop truck and pivot mast with load 90 degrees to right, then pivot mast back to straight ahead position.

(c) Deposit load to Station 2; back out; then travel forward to Station 3. Prior to engaging load at Station 3, raise mast to maximum lift height to activate relief valve. Lower forks and engage load.

(d) Pick up load at Station 3; back out; then travel forward and deposit load at Station 1, being sure all wheels travel over obstacles.

(e) Disengage load at Station 1; back out, being sure all wheels travel over obstacles; then travel forward to Station 2.

(f) Pick up load at Station 2; back out; then travel forward and before reaching Station 3, stop truck, pivot load 90 degrees to right, then pivot back to straight ahead position. Deposit load at Station 3.

(g) Steps 2 (b) through 2 (f) shall constitute one circuit. For 50 percent of the circuits, Station 2 and Station 3 shall be interchanged so that they are in clockwise position rather than in their initial counterclockwise position. Stations shall be changed at the end of each 8-hour operating day.

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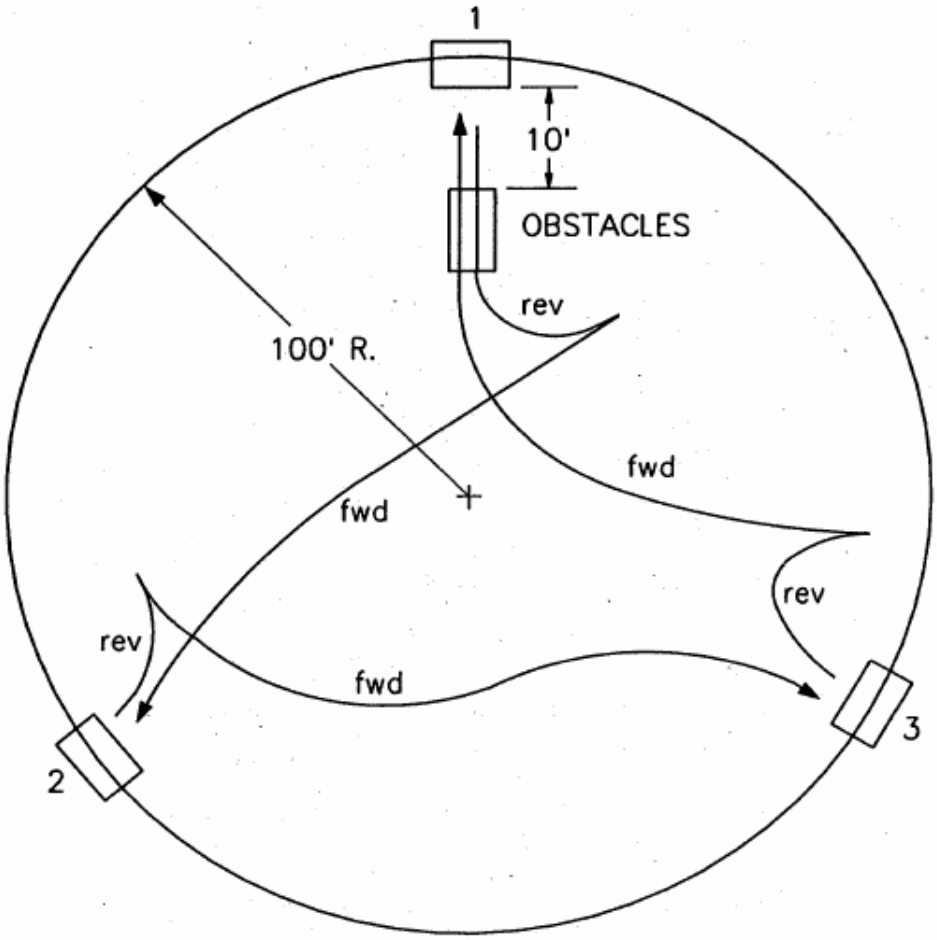
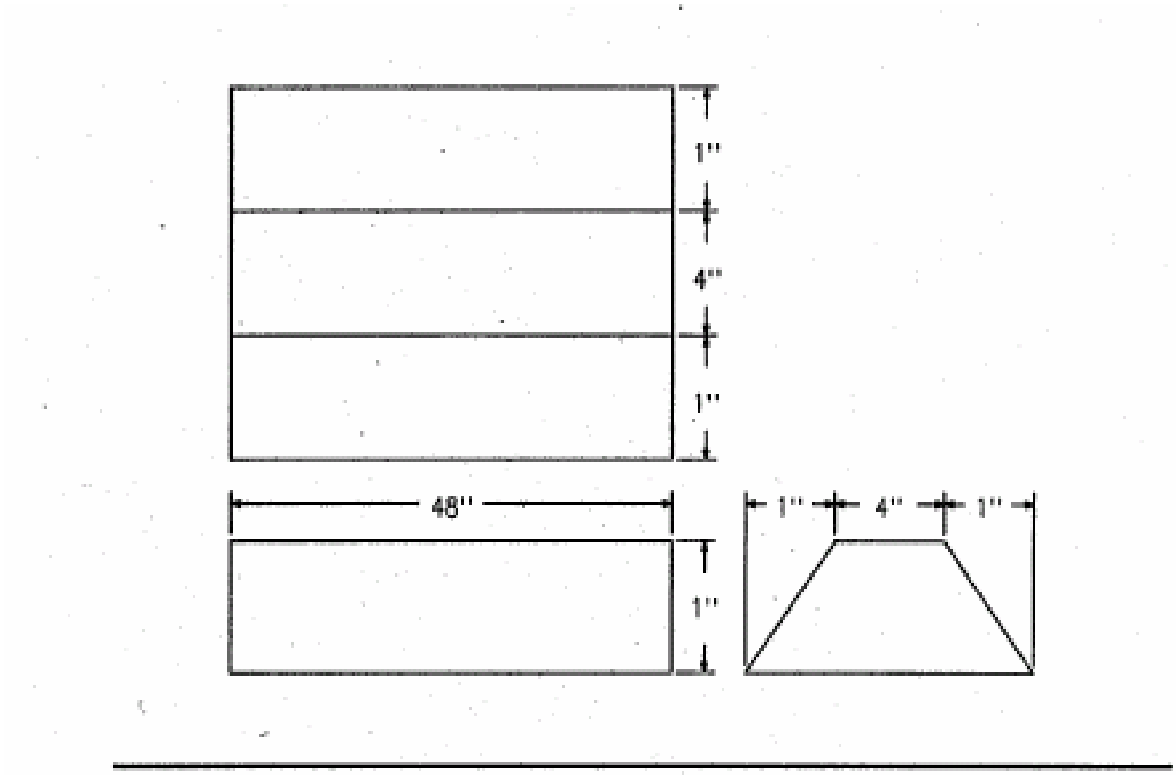


Figure A-1 TRUCK OPERATIONS COURSE

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CONSTRUCTION OF OBSTACLE S



OBSTACLE COURSE LAYOUT

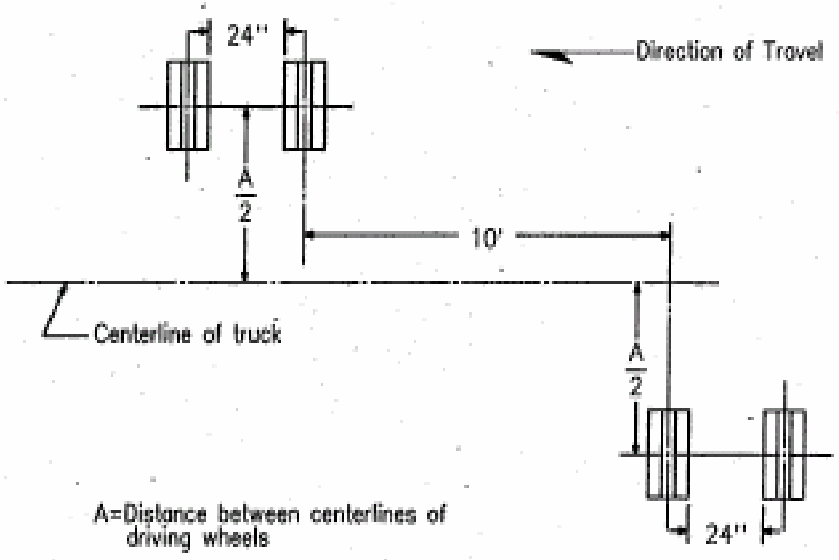


Figure A-2 OBSTACLES

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MILITARY INTERESTS:

Custodian:
Air Force - 99
Army - MI

CIVIL AGENCY
COORDINATING ACTIVITIES:
GSA-FSS

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

(Project 3930-2007- 003)

Note: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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